

Computer Networks

Lecture 14: Data Link Layer

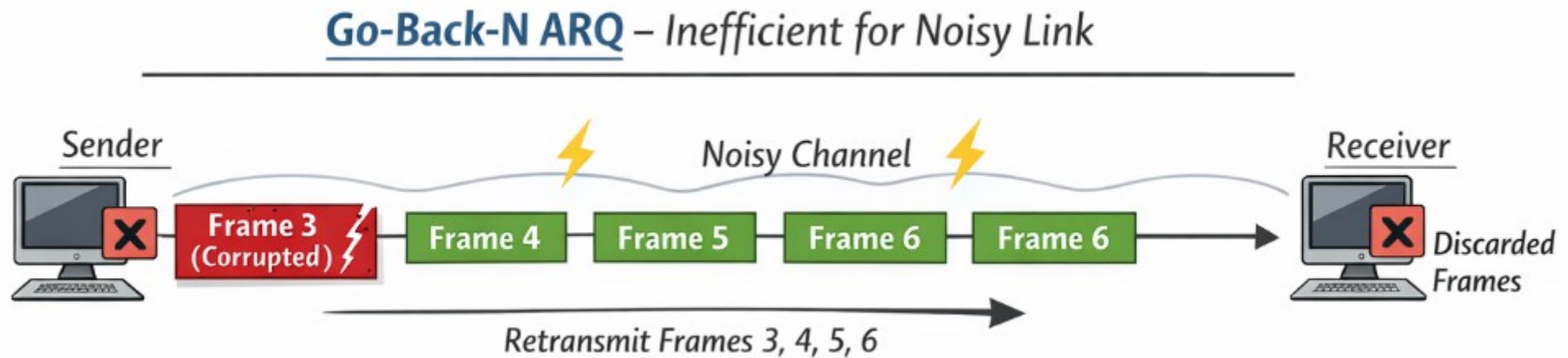
Content



- **Selective repeat Automatic Repeat Request**

Introduction

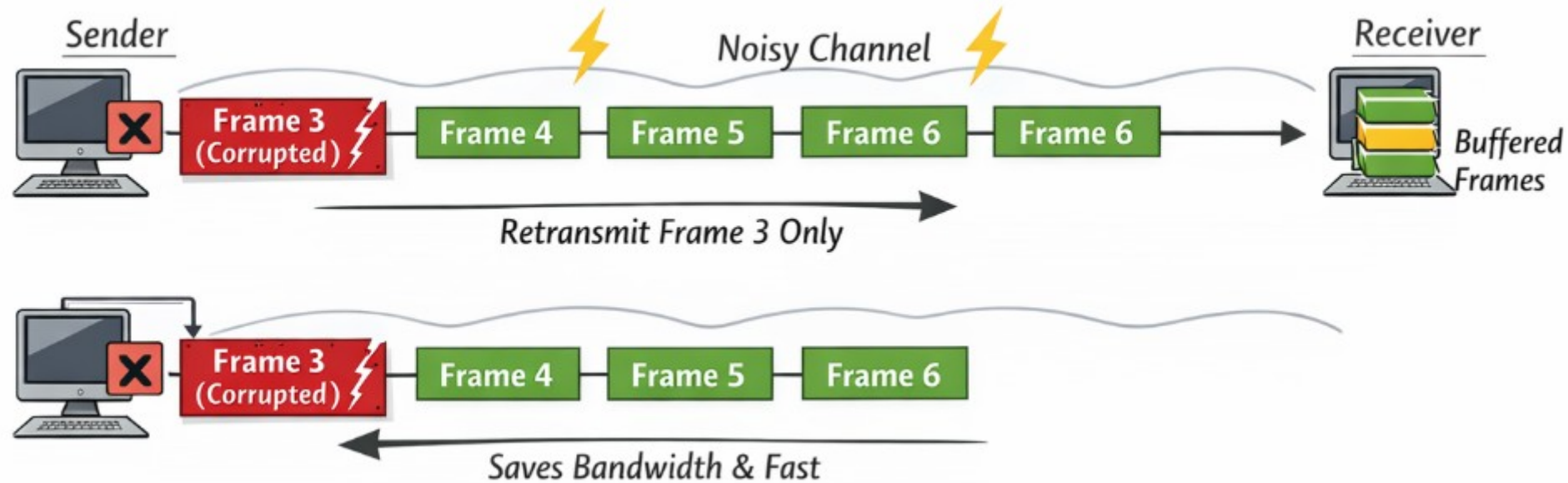
Go-Back-NARQ simplifies the process at the receiver site. The receiver keeps track of only one variable, and there is no need to buffer out-of-order frames; they are simply discarded.



However, this protocol is very inefficient for a noisy link. In a noisy link a frame has a higher probability of damage, which means the resending of multiple frames. This resending uses up the bandwidth and slows down the transmission.

Selective Repeat-ARQ

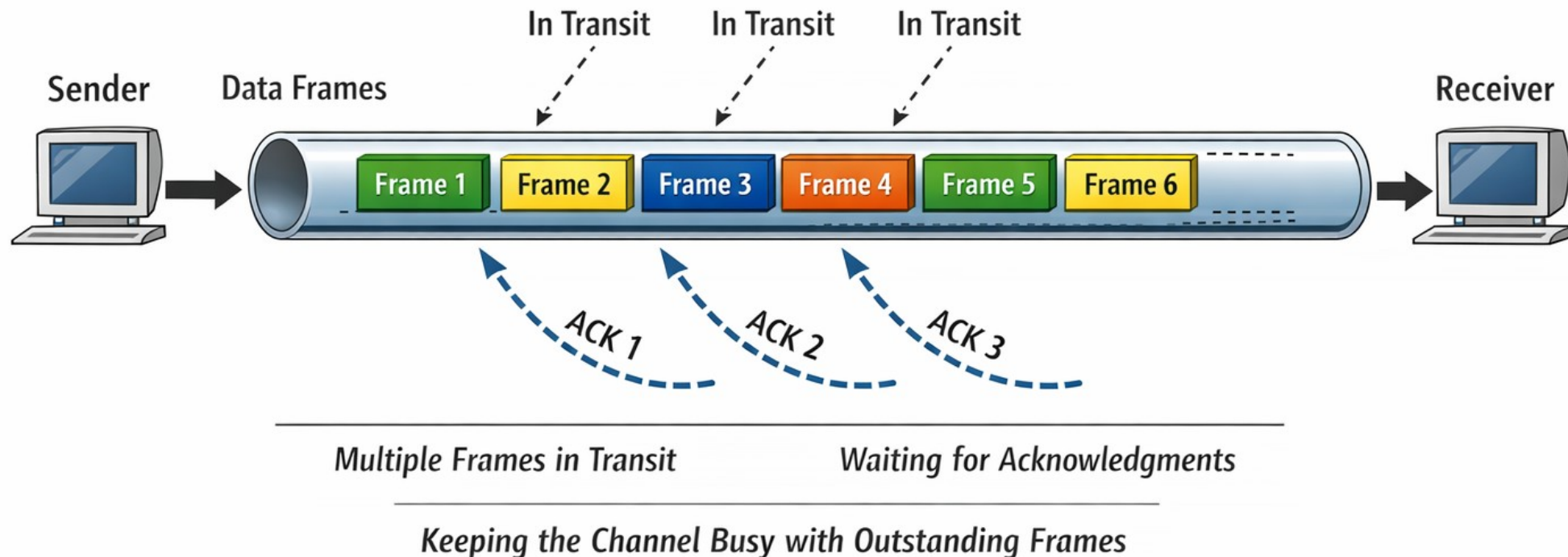
Selective Repeat Automatic Repeat request (SR ARQ) is an error-control protocol used in data communication to ensure reliable data transmission over unreliable channels.



In this approach the receiver can accept and buffer out-of-order frames and send individual acknowledgments for each correctly received frame.

Selective Repeat-ARQ

To improve the efficiency of transmission multiple frames must be in transition while waiting for acknowledgment. In other words, we need to let more than one frame be outstanding to keep the channel busy while the sender is waiting for acknowledgment



Selective Repeat-ARQ

Send Window

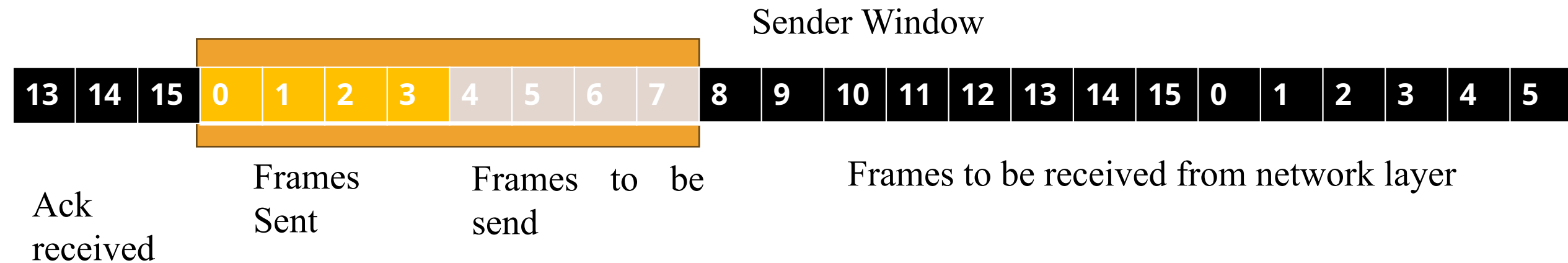
The send window is a logical (imaginary) range of sequence numbers. It represents which frames are allowed to be in transit

- The window covers a set of sequence numbers of data frames.
- Only frames whose sequence numbers lie inside the window can be transmitted.

The maximum size of the window is 2^{m-1}

Go-Back-N Automatic Repeat Request

Send Window



In Go-Back-N, the sender window is controlled using three variables:

Sf = Sequence number of the first outstanding (oldest unacknowledged) frame

Sn = Sequence number of the next frame to be sent

Ssize = Maximum number of frames allowed to be outstanding

Go-Back-N Automatic Repeat Request

Receive Window

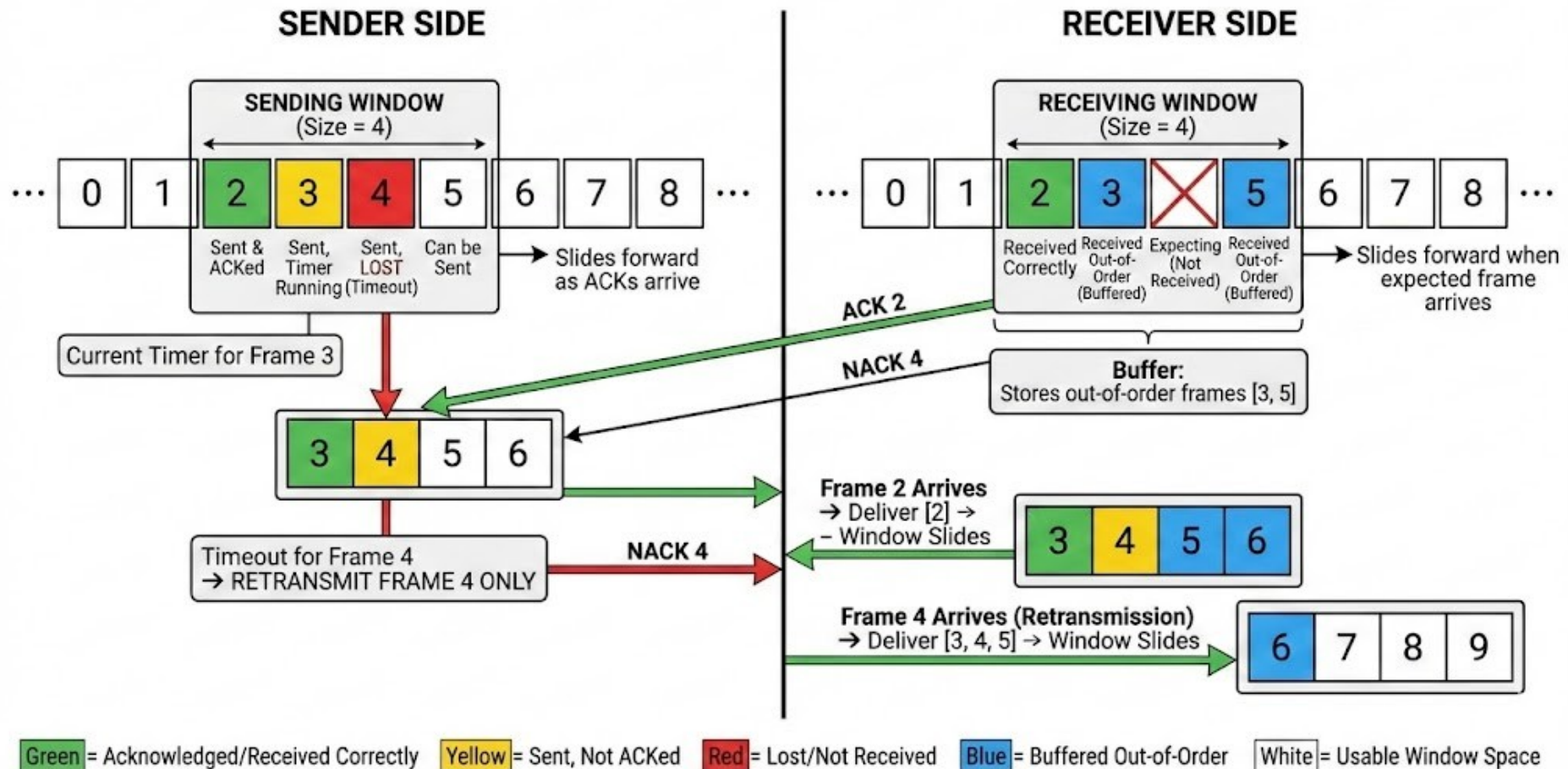
The size of the receive window is same as the size of the send window $2^m - 1$

Any frame arriving out of order is accepted



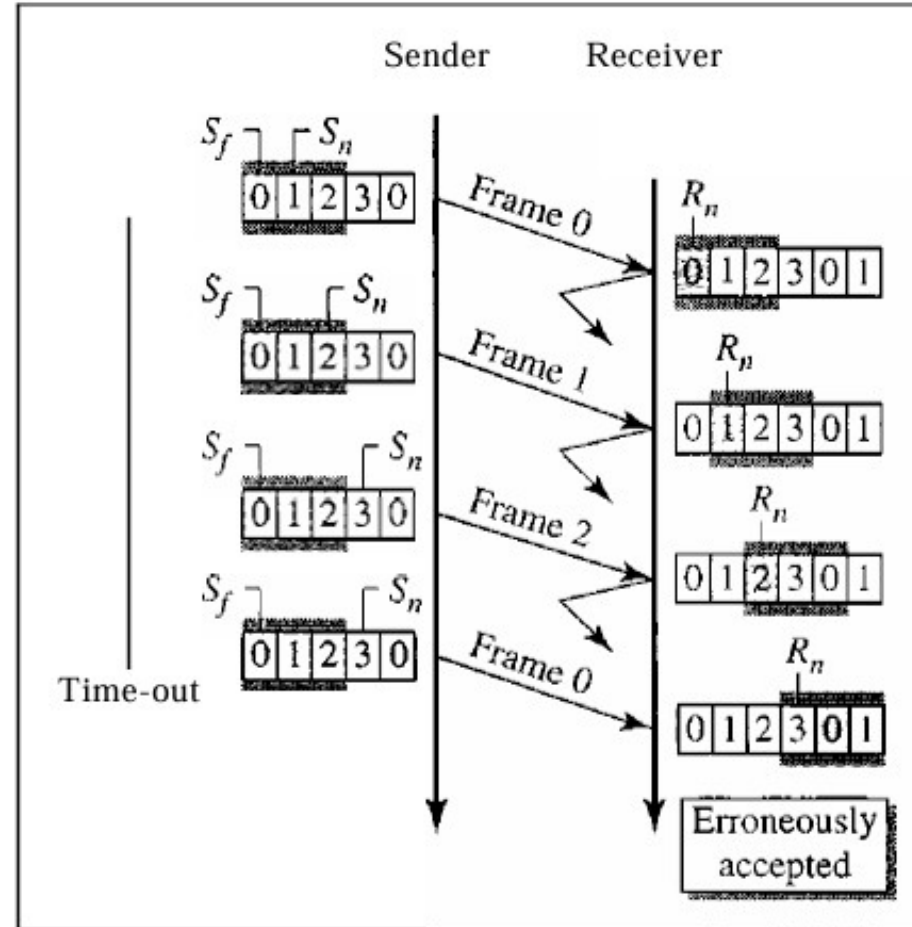
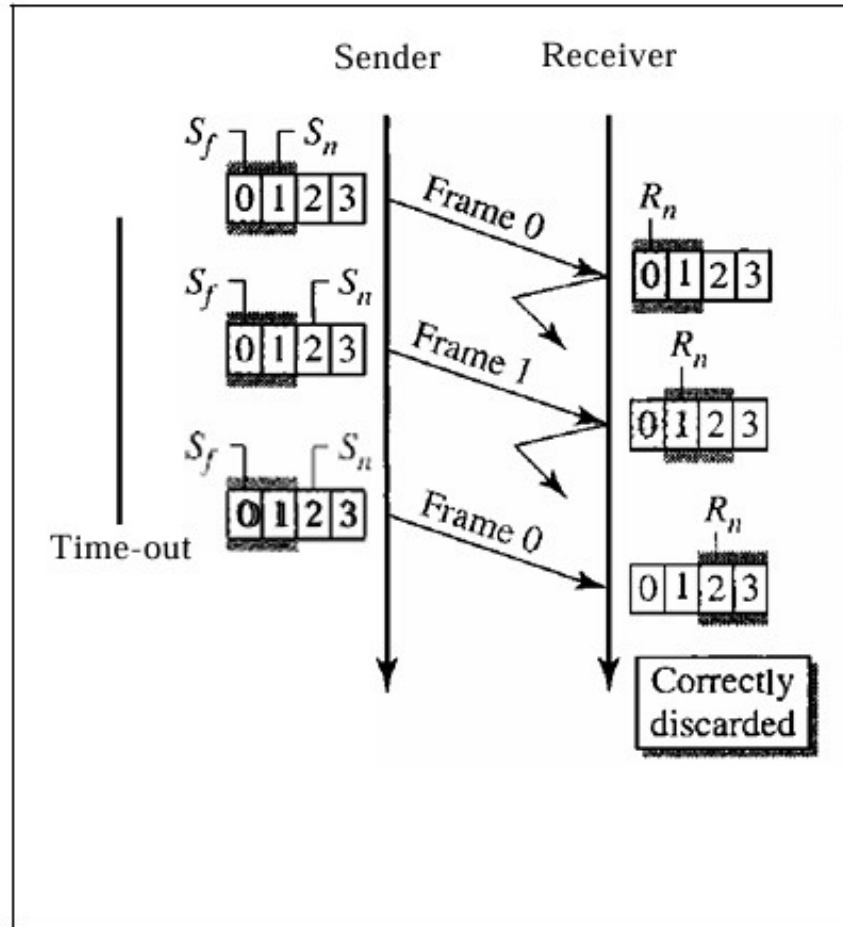
Go-Back-N Automatic Repeat Request

Window sliding

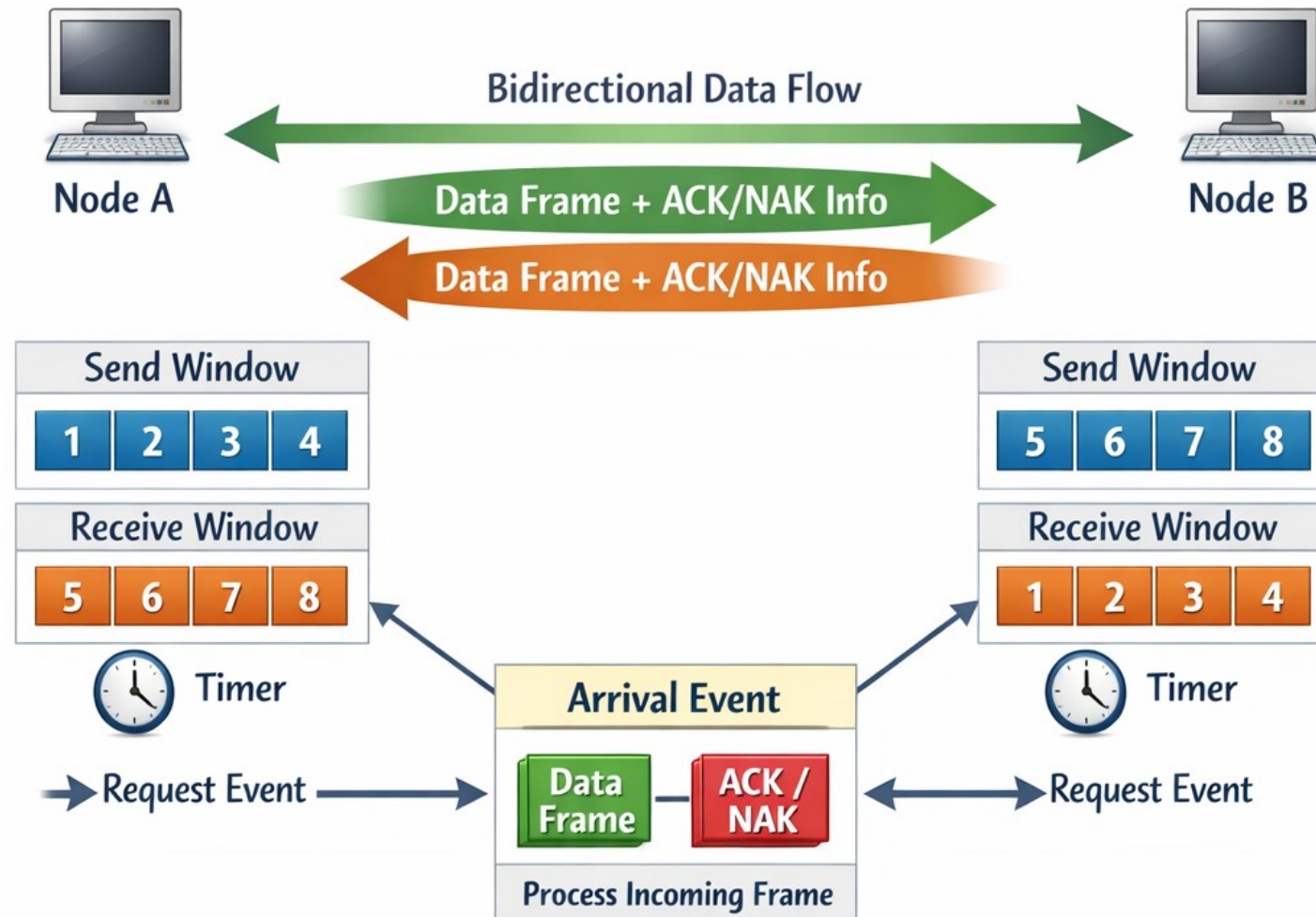


Go-Back-N Automatic Repeat Request

Why window size is $2^n - 1$



Piggybacking



Questions

1. If N is the size of window in selective repeat ARQ. Calculate the no. of sequence bits used.
Ans: $\log_2 N + 1$
2. If 5 is the no. of sequence bits used. Calculate the size of sender and receiver window in selective repeat ARQ and go back N ARQ
Ans: 16, 16
Ans: 31, 1
3. If bandwidth is 10 Mbps, the RTT is $1000 \mu s$ and frame size is 100 bits. Calculate the no. of sequence bits used in selective repeat ARQ to transmit the data in RTT.
Ans: 8
4. If M is the maximum sequence no. used in selective repeat ARQ. What is the window size.
Ans: $\frac{M+1}{2}$
5. If N is the total sequence no. used in selective repeat ARQ. What is the window size.
Ans: $\frac{N}{2}$

Questions

6. Using stop-and-wait ARQ protocol, the sender wants to transmit 10 data frames to the receiver. Out of these 10 data frames, every 4th data frame is lost. How many frames sender will have to send in total?
7. A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps. The size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 msec. Assuming no frame is lost, calculate the sender throughput in bytes per second.